CBSE Class 10 Mathematics

(Standard) Previous Year Question

Paper – Set 30/2/1

Time Allowed: 30 Minutes

Maximum Marks: 80

General Instructions

- All questions are compulsory.
- Internal choice is provided wherever applicable.
- Questions follow the current CBSE syllabus.

SECTION A – Objective Questions

1. If a quadratic equation has roots 2 and 3, what is the equation?

(a) $x^2 - 5x + 6 = 0$

(b) $x^2 + 5x + 6 = 0$

(c) $x^2 - 6x + 5 = 0$

(d) $x^2 + 6x + 5 = 0$

2. The value of sin 60° is:

(a) 1

(b) 0

(c) ½

(d) $\sqrt{3}/2$

3. Which of the following is irrational?

(a) 1/2

(b) 0.25

(c) $\sqrt{5}$

(d) 7

4. The sum of the first 10 natural numbers is:

(a) 100

(b) 45

(c) 55

(d) 50

5. If the distance between the points (1, 2) and (4, 6) is d, what is the value of d?

(a) 4

(b) 5

(c) $\sqrt{25}$

(d) 3

6. The number of tangents that can be drawn from a point outside a circle to the circle is:

(a) 0

(b) 2

(c) 1

(d) infinite

7. Which of the following statements is true?

(a) The sum of all interior angles of a triangle is 180°.

(b) The exterior angle of a triangle is greater than interior angle.

(c) Triangles do not have diagonals.

(d) All sides of an isosceles triangle are equal.

8. If the area of a circle is 154 cm², what is its radius?

(a) 7 cm

(b) 14 cm

(c) 5 cm

(d) 11 cm

9. The value of x that satisfies the equation $2x + 1 = 9$ is:

(a) 2

(b) 5

(c) 4

(d) 6

10. What is the common difference of the arithmetic progression 7, 10, 13, ...?

(a) 2

(b) 3

(c) 5

(d) 7

11. The coordinates of the midpoint of the line joining (2, 3) and (6, 7) are:

(a) (4, 5)

(b) (8, 10)

(c) (2, 7)

(d) (6, 3)

12. The HCF of 20 and 28 is:

(a) 4

(b) 2

(c) 8

(d) 12

13. The zeroes of the quadratic polynomial $x^2 - 4x + 4$ are:

(a) 2, 2

(b) −2, −2

(c) 0, 4

(d) −4, 0

14. If in ΔABC, AB = AC, it is called:

(a) Scalene triangle

(b) Isosceles triangle

(c) Right triangle

(d) Equilateral triangle

15. A line parallel to the base of a triangle divides the other two sides:

(a) Equally

(b) Unequally

(c) Proportionally

(d) Not at all

16. In a right-angled triangle, the side opposite to the right angle is called:

(a) Median

(b) Altitude

(c) Hypotenuse

(d) Base

17. The value of $3^2 - 4 \times 2 + 1$ is:

(a) 2

(b) 1

(c) 9

(d) 0

18. If $x = -1$ is a solution of quadratic equation $x^2 + 2x + 1 = 0$, then which statement is true?

(a) $x = -1$ is a double root

(b) $x = -1$ is not a root

(c) $x = -1$ is not possible

(d) $x = 1$ is a root

19. Assertion (A): The diagonals of a rectangle are equal.

Reason (R): A rectangle is a parallelogram.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation.

(c) A is true, R is false.

(d) Both A and R are false.

20. A die is thrown once. The probability of getting a number less than 5 is:

(a) 5/6

(b) 1/3

(c) 2/3

(d) 4/6